Paleomagnetism of the Middle Proterozoic Mount Nelson Formation: evidence for a regional remagnetization event in the Late Precambrian of the Cordillera

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Chemical and thermal demagnetization of 92 specimens from the Mount Nelson Formation (uppermost Purcell Supergroup) in the Purcell Mountains of southeastern British Columbia yields two distinct directional groups. MN-A, found by thermal and chemical demagnetization, has a mean direction of $D = 274°$, $I = 19°$ ($\alpha_{95} = 10°$, tilt corrected), which corresponds to a pole position at 156°E, 10°N. MN-A resides in fine-grained hematite and is similar to other overprint directions reported from middle and upper Belt–Purcell strata in the Glacier National Park – Clark Range area. Genesis of this overprint is related to some regional geological process, most probably the Goat River Orogeny (≃800–900 Ma). Direction MN-B, which is found after acid leaching of more than 250 h, has a mean direction of $D = 331°$, $I = 47°$ ($\alpha_{95} = +4°$, tilt corrected), corresponding to a pole at 119°E, 59°N. The MN-B pole is significantly different from all poles previously reported from the Belt–Purcell Supergroup (ca. 210°E, 20°S). As the age of the MN-B pole is unconstrained the significance of this marked discrepancy is at present uncertain.

La désaimantation chimique et thermique de 92 spécimens de la formation Mount Nelson (supergroupe Purcell tardi-supérieur), des chaînons Purcell du sud-est de la Colombie-Britannique, fournit deux groupes distincts de direction. La désaimantation thermique et chimique définit un groupe MN-A de direction moyenne $D = 274°$, $I = 19°$ ($\alpha_{95} = 10°$ après correction pour pendage) qui indique une position du pôle à 156°E, 10°N. L'aimantation du groupe MN-A est portée par l'hématite à grain fin et est analogue aux directions de surimpression rapportées pour les strates du Belt–Purcell moyen et supérieur dans le Parc National des Glaciers – région du chaînon Clark. L'origine de cette surimpression est en relation avec des processus géologiques régionaux, plus probablement avec l'orogénèse Goat River (≃800–900 Ma). La direction du groupe MN-B, laquelle fut révélée après une période de lessivage à l'acide de plus de 250 h, possède une valeur moyenne de $D = 331°$, $I = 47°$ ($\alpha_{95} = +4°$ après correction pour pendage), et correspond à un pôle situé à 119°E, 59°N. Le pôle MN-B diffère considérablement de tous les pôles rapportés jusqu'à maintenant pour le supergroupe Belt–Purcell (ca. 210°E, 20°S). Vu que l'âge du pôle MN-B n'est pas déterminé, l'importance de cette position différente du pôle ne peut être appréciée en ce moment.
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Introduction

Strata of the Middle Proterozoic Belt–Purcell Supergroup[1] exposed in the Purcell Mountains of southern British Columbia and adjacent Idaho and Montana were uplifted, block faulted, folded, and metamorphosed prior to deposition of the Upper Proterozoic Windermere Supergroup (Leech 1962). Metamorphic fabrics, intrusive relationships, isotopic age determinations, and paleomagnetic poles indicate that this deformation resulted from two distinct episodes of tectonism (McMechan and Price 1982). The older episode (≃1300–1350 Ma; East Kootenay Orogeny) apparently coincided with the termination of Belt–Purcell sedimentation and involved folding, regional metamorphism, and, locally, granitic intrusion. The younger episode (≃800–900 Ma; Goat River Orogeny) occurred during initial deposition of the Windermere Supergroup and involved uplift, block faulting, and low-grade regional metamorphism (McMechan and Price 1982). Similar tectonic episodes (Racklan and Hayhook Orogenies) have recently been recognized in Proterozoic strata exposed in the northern Canadian Cordillera (Eisbacher 1978; Young *et al.* 1979).

The stratigraphically highest unit of the Purcell Supergroup exposed in southeastern British Columbia is the Mount Nelson Formation (Table 1), a sequence of quartzite, dolomite, and argillite. It is unconformably overlain by Upper Proterozoic conglomerates of the Toby Formation. In an attempt to clarify the significance of this unconformity a paleomagnetic study of the Mount Nelson and Toby Formations was undertaken. A total of 92 samples from 14 sites in the Mount Nelson Formation and 120 samples from 20 sites in the Toby Formation were collected where these formations outcrop above Toby Creek, near Invermere, British Colum-

[1]The term Belt–Purcell Supergroup is derived from the proper names Belt Supergroup and Purcell Supergroup for Middle Proterozoic strata exposed in the northern United States and southern Canada, respectively.

TABLE 1. Lithostratigraphic correlation chart, Purcell Supergroup, Clark Range, and northern Purcell Mountain areas (after McMechan 1981)

	N. PURCELL MOUNTAINS	CLARK RANGE
MIDDLE PROTEROZOIC PURCELL SUPERGROUP	TOBY FM.	CAMBRIAN
	MT. NELSON FM.	ROOSVILLE FM.
	DUTCH CREEK FM.	PHILLIPS FM.
		GATEWAY FM.
		SHEPPARD FM.
		PURCELL LAVA
	VAN CREEK FM.	SIYEH FM.
	KITCHENER FM.	
	CRESTON FM.	GRINNELL FM.
		APPEKUNNY FM.
	UPPER UNIT	ALTYN FM.
		WATERTON FM.
		"FM. 6"
	ALDRIDGE FM.	"FM. 5"
		"FM. 4"
		"FM. 3"
		"FM. 2"
		"FM. 1"
	base not exposed	

bia (Fig. 1). The samples from the Toby Formation did not yield coherent remanence directions. Those from the Mount Nelson Formation yielded two remanence components. In this paper, we first outline the characteristics of the individual remanence components that can be recognized in the Mount Nelson data set, then discuss the paleomagnetic evidence for the relative ages of the individual components. By comparison of these new results with previously published results of Late Precambrian and early Paleozoic age, the age and significance of the individual components are evaluated.

Paleomagnetic methodology

All specimens were subjected to individual demagnetization treatments. No bulk cleaning was performed. Remanence measurements and thermal treatments of the specimens were made on Schonstedt instruments (DSM and TSD). Thermal demagnetization steps were usually 50°C between 200 and 550°C, then 10°C up to a maximum of 690°C. Acid leaching was also used as a cleaning technique. In this process, a specimen is immersed in 12 *N* HCl for a specified period of time. The sample was then washed, dried in quasi-zero magnetic

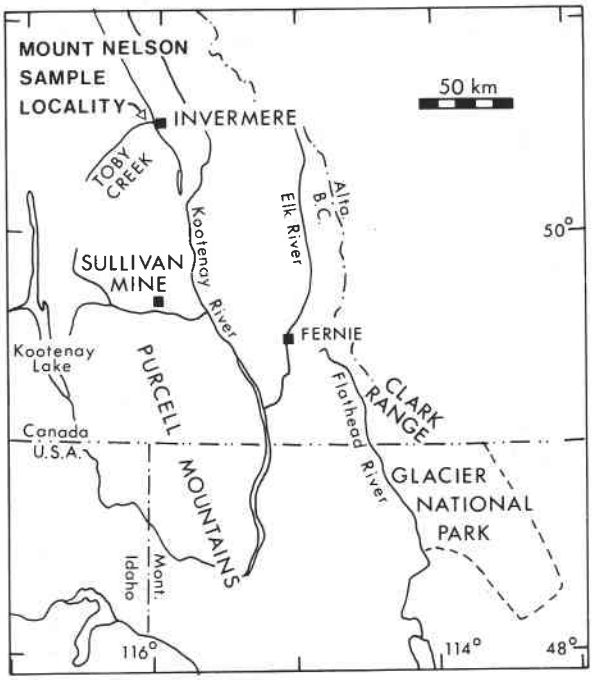


FIG. 1. Index map showing location of areas discussed in text.

conditions, and measured. The procedure was repeated until no detectable remanence remained. To promote leaching the specimens were overcored as suggested by Morris and Carmichael (1978). During the initial stages of this treatment, 30–50 h intervals of acid immersion were used between measurements; as the experiment progressed, these individual periods of leaching were extended up to a maximum of 500 h. The maximum accumulated leaching time for certain specimens was as much as 6350 h.

As both thermal and acid leaching techniques gave very well defined directional data sets, it was not necessary to employ any of the available vector analysis techniques for the separation of multicomponent magnetization, although orthogonal demagnetization plots of each specimen were prepared as a check for the presence of possible multicomponents. The following hierarchical system was used in calculating the representative directions: specimens combined to give sample means; sample means combined to give the overall mean. Standard Fisher (1953) paleomagnetic statistics were used at all levels of this hierarchy.

Results

Only two directions were detected in the Mount Nelson data set (Table 2). One, MN-A, was found during thermal demagnetization and the other, MN-B, was detected only by acid leaching.

Upon thermal treatment, all specimens had almost

TABLE 2. Summary of paleomagnetic results, Mount Nelson Formation

		N	R	k	α_{95}°	D°	I°
A	*In situ*	14 sites	13.30	19	9	268	22
A	Tilt corrected	14 sites	13.27	18	10	274	19
A	*In situ*	92 specimens	81.31	8	5	272	28
A	Tilt corrected	92 specimens	80.96	8	5	280	23
B	*In situ*	20 specimens	19.72	69	4	320	61
B	Tilt corrected	20 specimens	19.75	75	4	331	47

NOTES: R, k, and α_{95} are the standard Fisher paleomagnetic statistics. D and I, are, respectively, the mean remanence declination and inclination.

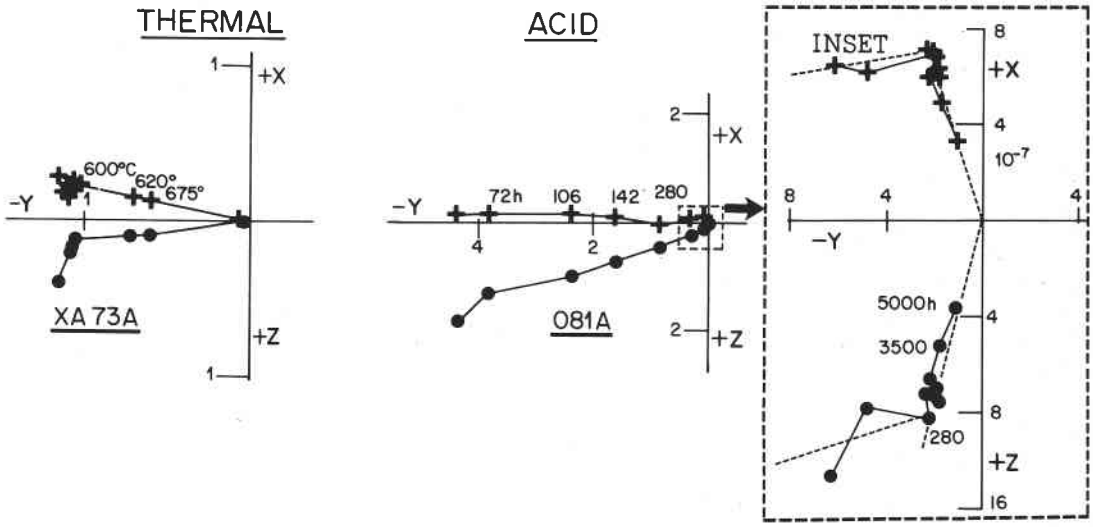


FIG. 2. Orthogonal vector projections showing examples of the demagnetization characteristics of samples from the Mount Nelson Formation. Upon thermal treatment, a univectorial remanence component is observed that only begins to decay above 600°C. Upon acid leaching, the westerly shallow vector is removed during the first 300 h of treatment, and then the northwest intermediate vector is found. In the orthogonal vector projections, crosses represent projections onto the horizontal (XY) plane, and dots represent projections onto either of the two vertical planes (ZY, ZX). The adopted convention is $+X$ is north, $+Y$ is east, and $+Z$ is down.

identical remanence characteristics. Throughout the treatment, the demagnetization trajectories were univectorial; there was no trace of any other remanence component. Few specimens showed any significant intensity decay below thermal treatment levels of 600–620°C, and there was very little evidence of any remanence residing in magnetite. Within the temperature range 620–670°C, the decay of intensity was still limited, and most specimens retained up to 50% of their natural remanent magnetization (NRM) even after treatment up to this temperature. The small decay in this interval, however, is sufficient to define linear segments on the orthogonal diagrams (Fig. 2). All the magnetic characteristics of this remanence component indicate that fine-grained pure hematite was the main remanence-carrying phase.

The directions found from these thermal demagnetization experiments have well defined westerly declinations. The inclination (tilt corrected) varies both between and within sites from −5 to +30°. As shown in Fig. 3*a*, there is a definite suggestion of a meridional distribution in the tilt-corrected data set. There are two possible explanations for this feature of the data set. Either it is an apparent feature of the data set resulting from inaccurate tilt correction, or it is a real feature possibly reflecting prolonged acquisition of this remanence component. The former explanation appears unlikely because of the following. (a) A similar distribution is also apparent although less well defined in the *in situ* data set. (b) Dips in this area are low, generally <25°, and fold axes have little plunge. It is therefore, impossible for tilt-correction errors to cause 30° differences in inclination. (c) The 30°+ variation in magnetic inclination is observed within a single site, over which

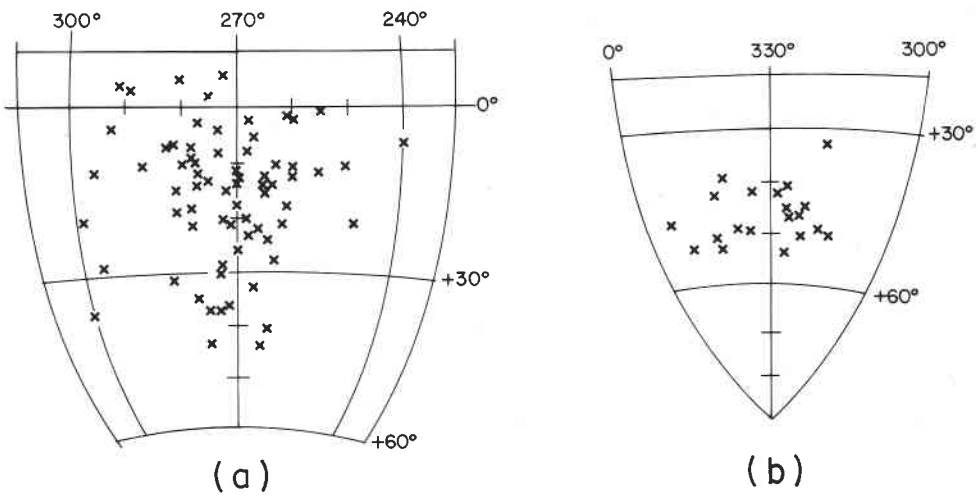


FIG. 3. Equatorial equal-area stereographic projections of the individual specimen remanence components found in this survey of the Mount Nelson Formation. (*a*) Cleaned A direction, tilt corrected, (*b*) Cleaned B direction, tilt corrected. The younger MN-A direction exhibits a distinct scatter in magnetic inclination, whereas MN-B appears to have a more Fisherian distribution.

there is no significant structural variation. The latter explanation in terms of prolonged remanence acquisition is, therefore, preferred. Assuming a simple Fisherian distribution (not strictly valid) for the MN-A directions, the calculated mean direction (tilt corrected) for this set is $D = 274°$, $I = +19°$, ($N = 14$ sites, $\alpha_{95} = 9°$). This corresponds to a paleomagnetic pole at 156°E, 10°N. Tilt correction causes only a very minor reduction in the Fisher precision parameter; hence it is not possible to use the fold test as a discriminant of remanence age. Note that the *in situ* pole position (162°E, 8°N) is not markedly different from the tilt-corrected pole.

The MN-B direction was only found after acid leaching of the specimens. During the first 250–350 h of leaching, the MN-A direction was removed, and the intensity rapidly decayed (Fig. 2). Beyond 350 h, the vector changed suddenly over to the MN-B direction. Usually at this point, less than 1% of the original NRM intensity remained. After as much as 6350 h of total leaching time for a specimen, no other vector components (Fig. 2) were detected and it was only after 2000–3000 h of leaching that any intensity decay was observed in the MN-B vector. Only 20 specimens were treated by acid leaching because at the time the initial thermal experiments were performed it was not apparent that acid leaching would reveal the MN-B remanence component.

As shown in Fig. 3*b*, the MN-B directions obtained after leaching do not show a broad longitudinal distribution as found in the MN-A directions. The tilt-corrected mean direction for this data set is $D = 331°$, $I = +47°$ ($N = 20$ specimens, $\alpha_{95} = 4°$). This corresponds to a paleomagnetic pole position at 113°E, 59°N. Tilt correction in this case produces a small increase in Fisher precision but this increase is not sufficient to constitute a positive fold test of remanence age.

There are no geological tests of remanence age applicable to this data set. Hence, there is no definitive evidence of either the absolute, or relative, ages of the two magnetic vectors recognized in the Mount Nelson Formation. On the basis of the chemical leaching and thermal treatment experiments, it is certain that the MN-A direction resides in hematite that occurs as a fine-grained coating on lithic grains and as cement. The MN-A direction records a phase of hematite deposition (crystallization) that postdates the initial sedimentation. The MN-B direction is probably related to the presence of coarser grained particles (possibly hematite or magnetite of probable detrital origin). Two possible extreme views on the age of this component are: (a) if magnetically very stable then MN-B may be recording an early Belt deposition related pole; or (b) if magnetically unstable then MN-B may be recording a more recent magnetic resetting, for example during the Laramide Orogeny.

Discussion

Early paleomagnetic surveys from the Belt–Purcell Supergroup have been interpreted to indicate that the continent occupied a nearly constant position relative to the pole during deposition of the whole supergroup (Collinson and Runcorn 1960; Black 1963). More recent surveys (in some instances using only limited remanence cleaning (thermal demagnetization) and vector analysis techniques) have for the most part confirmed the univectorial results of the earlier studies (Evans *et al.* 1975; Vitorello and Van der Voo 1977; Elston and Bressler 1980).

However, other magnetic vectors are present at least locally in these strata. In the Glacier National Park area of Montana, Vitorello and Van der Voo (1977) separated a component with intermediate unblocking temperatures, with mainly westerly declinations, from samples of the Spokane Formation (middle Belt Supergroup equivalent to the Grinnell). They also recognized the presence of similar magnetizations in middle and upper Purcell strata [Grinnell = Spokane to Phillips (Kintla C) Formations] from the Clark Range, in data previously published by Black (1963), Runcorn (1964), and Evans *et al.* (1975). This overprint vector, in the Glacier National Park – Clark Range area, denoted herein by GNC-B, is identical to the MN-A direction that is so prominent in the present study. In addition, both MN-A and GNC-B components are thought to be residing in hematite, and in both the directions are spread in a non-Fisherian distribution (see Fig. 5 of Vitorello and Van der Voo 1977). The only essential difference between the results from the Glacier National Park – Clark Range area and the Mount Nelson Formation is the extent to which this magnetic signature is developed. In middle and upper Belt–Purcell strata of the Glacier National Park – Clark Range area, the vector is seen as a partial overprint, whereas in the Mount Nelson Formation this vector dominates the whole blocking-temperature spectrum. From this distribution it may be concluded that whatever process gave rise to this signature was more effective in the Purcell Mountains than in the Glacier National Park – Clark Range area.

The MN-A pole (156°E, 10°N) and the secondary GNC-B pole (163°E, 5°N) of Vitorello and Van der Voo (1977) are similar to both Late Proterozoic (≃850 Ma; Berger and York 1980) and Middle Cambrian poles (e.g., Watts *et al.* 1980*a*,*b*; Dankers and Lapointe 1981) from elsewhere in North America. There is no evidence of a Middle Cambrian thermal and (or) deformational event in either the Purcell Mountains or Glacier National Park – Clark Range areas, and it is improbable that the MN-A and GNC-B poles record magnetizations acquired during the Middle Cambrian. Both poles, however, can be related to thermal overprinting produced by anomalously high heat flow associated with the uplift, block faulting, and probable rifting (Stewart 1972) that accompanied Windermere sedimentation (Goat River Orogeny; Vitorello and Van der Voo 1977). The main deformation accompanying this event occurred in the Purcell Mountains, west of the Glacier National Park – Clark Range area. This is consistent with the paleomagnetic results, which suggest the remagnetization process was more effective in the Purcell Mountains than in the Glacier National Park – Clark Range area.

Most poles from the Belt–Purcell Supergroup, including those from approximately the same stratigraphic level as the Mount Nelson Formation, plot around 210°E, 20°S (see Elston and Bressler 1980, Fig. 7). Conventionally, these poles have been considered as recording the initial magnetizations in these units. The MN-B pole reported here from the Mount Nelson Formation is widely divergent from this group. However, it is the first pole determination from the Belt–Purcell Supergroup that is based on acid leaching. Unfortunately, its time of acquisition could not be determined; hence its geological significance is uncertain. Clearly, confirmation of this pole and better determination of its time of acquisition are required.

Of especial importance is the necessity of establishing the age of the MN-B pole: if the MN-B pole dates from the time of sedimentation then this implies that at least some of the common belt poles are not original but reset secondary magnetizations. Fedikow (1978) on the basis of a failed fold test, argued for such a postdeformation remanence acquisition event in the lower Purcell Aldridge Formation near the Sullivan mine. Similarly, there are geological reasons to suggest that the Belt–Purcell poles may have been reset during the East Kootenay Orogeny. The Belt–Purcell Supergroup forms a northeasterly tapering wedge of predominantly fine-grained clastic and impure carbonate rocks up to 20 km thick (Harrison 1972) that unconformably overlaps Precambrian basement rocks of the North American craton (Peele 1893; Weed 1899; Lowell 1956; McMannis 1963). This wedge appears to be a prograded continental terrace wedge, because a lower turbidite sequence at least 7 km thick grades upward into tidal-flat, flood-plain, and carbonate shelf deposits (Price 1964). Price (1964) further suggested that the northern Gulf of Mexico may be a modern analogue to the Belt–Purcell sequence. An estimate of the average sedimentation rate for the Mesozoic–Cenozoic Gulf Coast is 0.15 m/1000 years (Schwab 1976), which is considerably higher than Schwab's average rate for modern continental terraces of 0.035 m/1000 years. If the relatively rapid average rate of sedimentation for the Gulf Coast region is assumed for the Belt–Purcell sequence, then the 20 km of sediment would have taken approximately 130 Ma to be deposited. However, the presently reported poles from the Belt–Purcell Supergroup, when compared with the North American polar wandering curve for the Middle Proterozoic (e.g., Irving 1979), suggest deposition of the entire supergroup within a period of less than 40 Ma. This would require rapid average rates of sedimentation of 0.5 m/1000 years. Such rates can occur for a few millions of years in foreland, rift, and intermontane basins (Fischer 1969; Spencer 1974; Schwab 1976), in large delta complexes (Boyce 1973), and in carbonate buildups

(reefs) (Mountjoy 1975, 1980), but they appear to be unrealistic for the carbonate shelf and intertidal deposits that constitute much of the Belt–Purcell Supergroup.

Clearly there is an urgent need for detailed geological tests of the age of remanence acquisition in Belt–Purcell strata for distinguishing between the two explanations of the apparently constant Belt pole position—(a) a clastic wedge that was formed by apparently high deposition rates, or (b) a regional remagnetization event that reset poles in the lower Belt–Purcell sediments during metamorphism associated with the East Kootenay Orogeny.

Conclusions

Thermal and chemical leaching treatments of specimens from the Mount Nelson Formation in the Purcell Mountains of southeastern British Columbia yielded two distinct remanence directions.

Pole MN-B, derived by acid leaching, yields a pole at 113°E, 59°N. This pole is divergent from all previously reported poles from the Belt–Purcell Supergroup. The significance of this divergence is not clear.

Pole MN-A (156°E, 10°N) is identical to secondary poles previously reported by Vitorello and Van der Voo (1977) from middle and upper Belt–Purcell formations in the Glacier National Park – Clark Range area. Thermal and chemical leaching experiments on samples from the Mount Nelson Formation indicate that the MN-A direction is related to hematite deposition and remanence blocking that probably took place over a geologically significant period of time. The remagnetization event was more severe in the Purcell Mountains than in the Glacier National Park area and was probably produced by anomalously high heat flow associated with the Goat River Orogeny, dated at approximately 800–900 Ma.

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